HireSmart ATS — AI-Based Resume Screening & Feedback Platform

Full-Stack AI & Machine Learning Project

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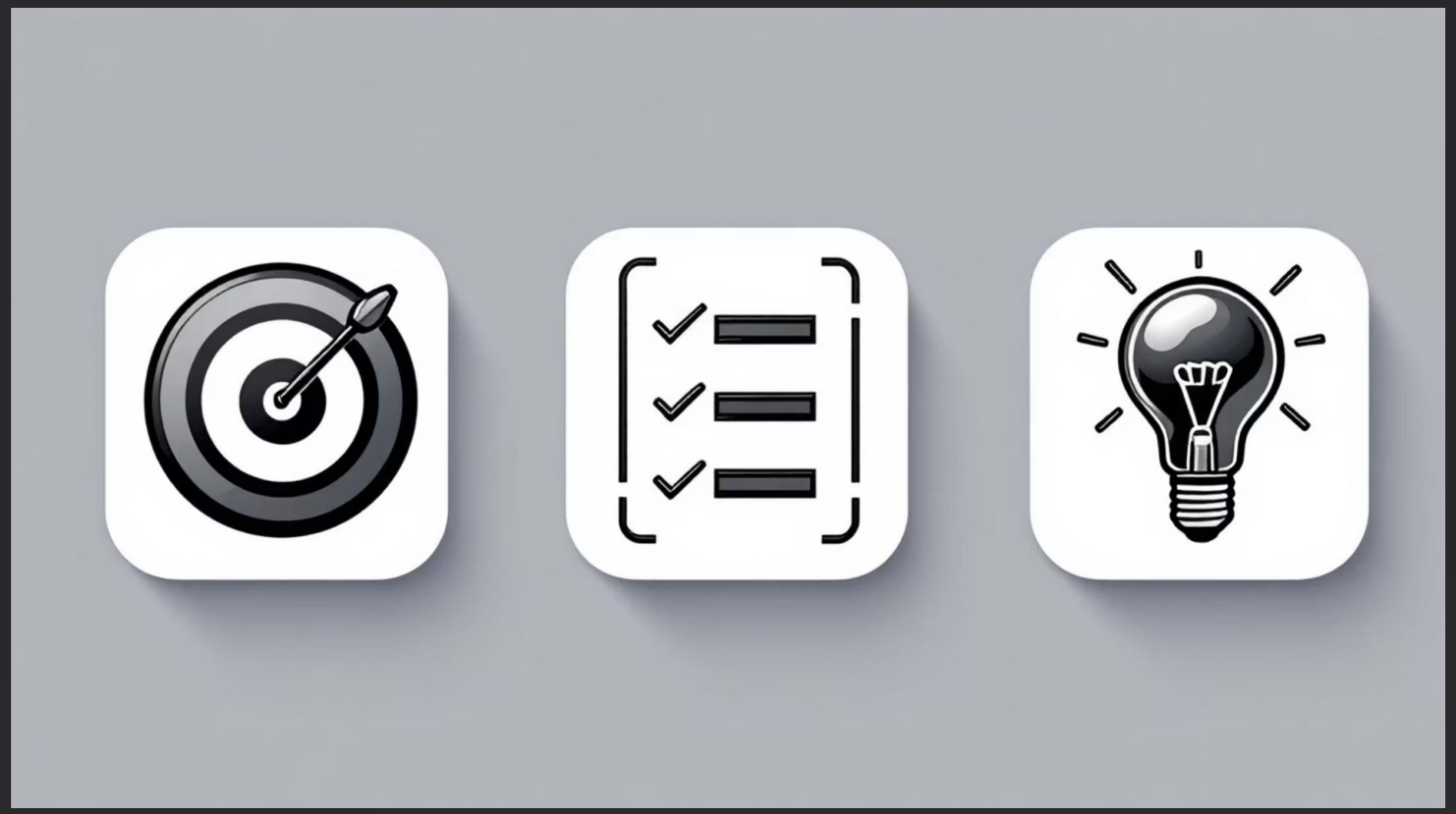
Problem Statement

- **What is an ATS?** Applicant Tracking Systems filter resumes at scale, often before a human reviews them.
- **Why many resumes get rejected:** format issues, missing keywords, poor parsing, and weak structure.
- **Need:** An intelligent evaluation that explains rejections and guides improvements.



Project Objectives

- Build an **AI-powered resume screening** system that mirrors ATS behavior.
- Provide **JD-based analysis** and an **ML-based general score** for resumes.
- Deliver actionable feedback so users can optimize resumes for ATS parsing and recruiters.

System Overview

High-level architecture: Frontend (UI) ⇔ Backend API ⇔ ML Model.
Data flow: PDF upload → parsing → feature extraction → scoring & suggestions → results displayed.

Frontend

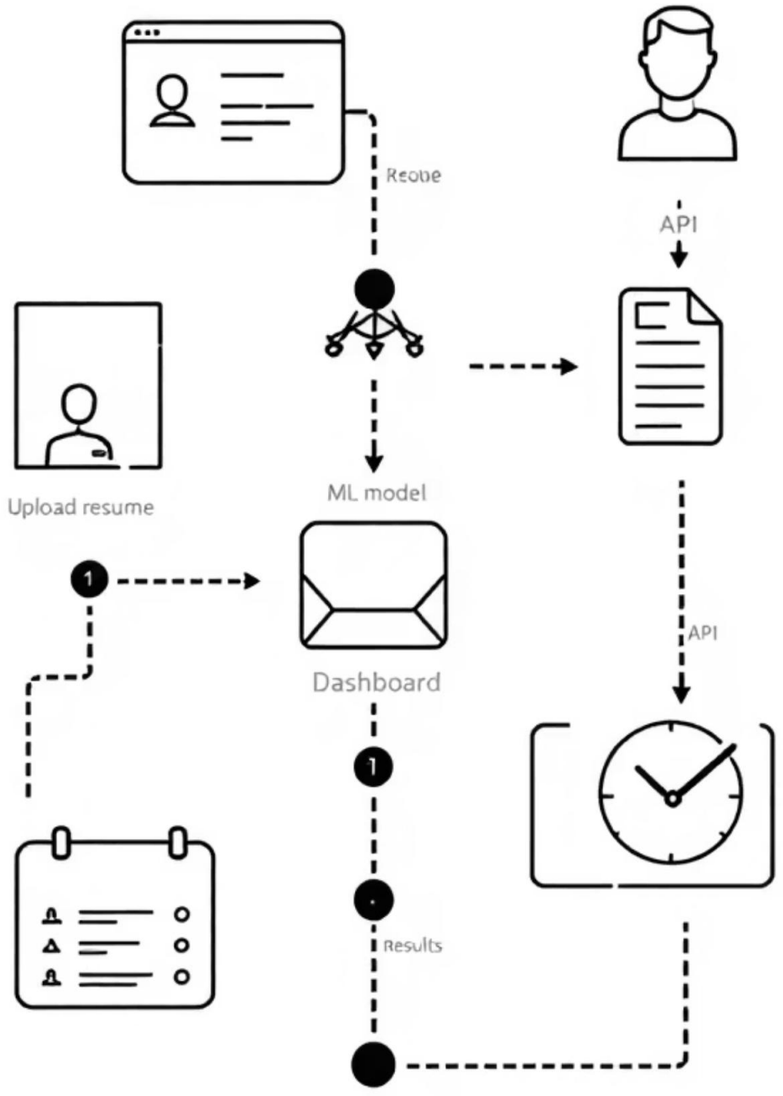
React app for uploads, visualizations, and feedback

Backend API

FastAPI endpoints handle parsing and prediction

ML Model

TF-IDF features + regression for continuous scoring



Key Features

Resume & JD Upload

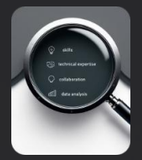
Accepts PDF resumes and job descriptions for comparative analysis.

ATS Match Score

JD-specific match percentage highlighting missing keywords.

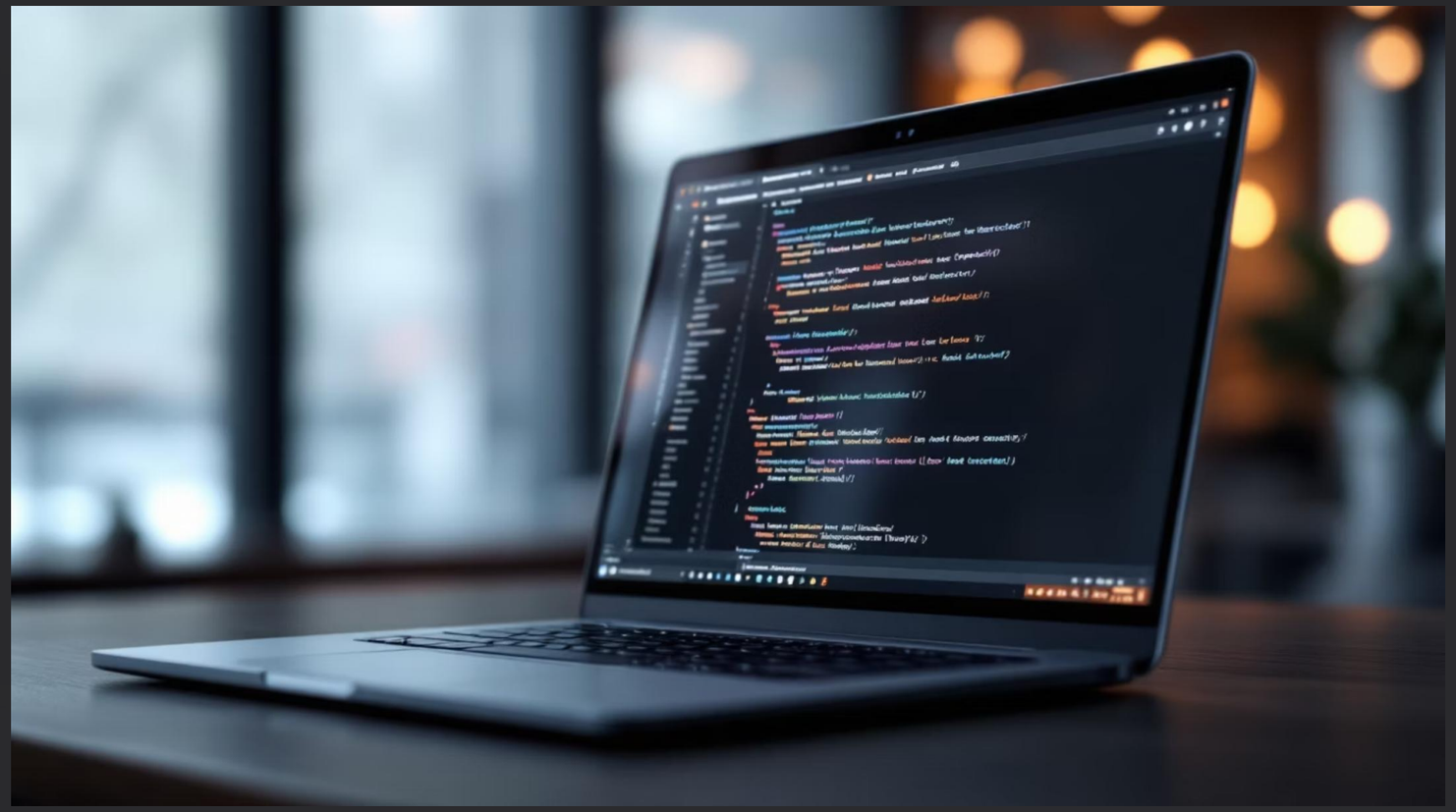
Resume Rating

Structured rating covering format, sections, and readability.



Skill Gap Analysis

Detects missing skills and offers phrasing suggestions for ATS parsing.



Technology Stack

Frontend

React · TypeScript · Tailwind CSS – responsive UI & file handling

Backend

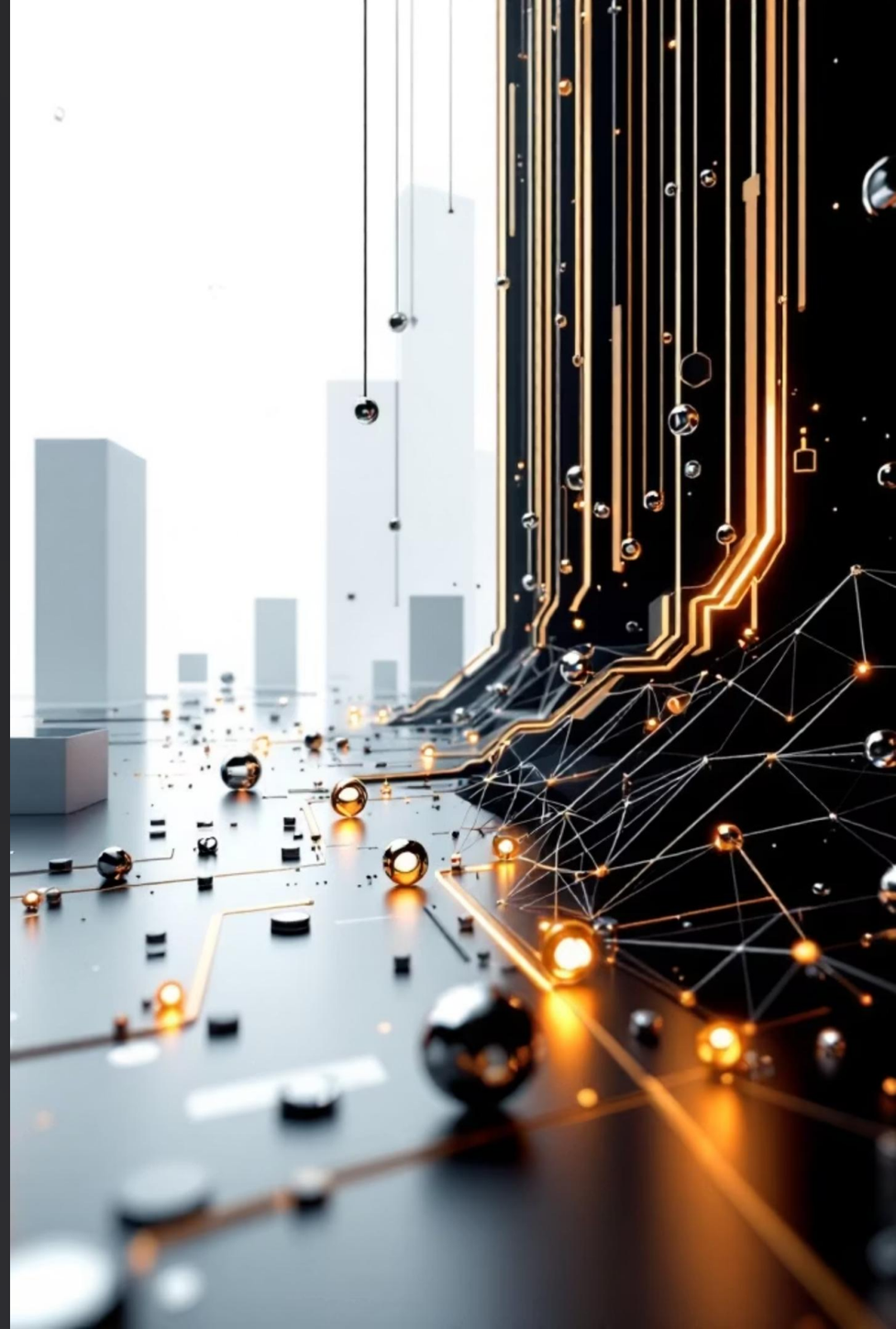
Python · FastAPI – scalable API endpoints and model serving

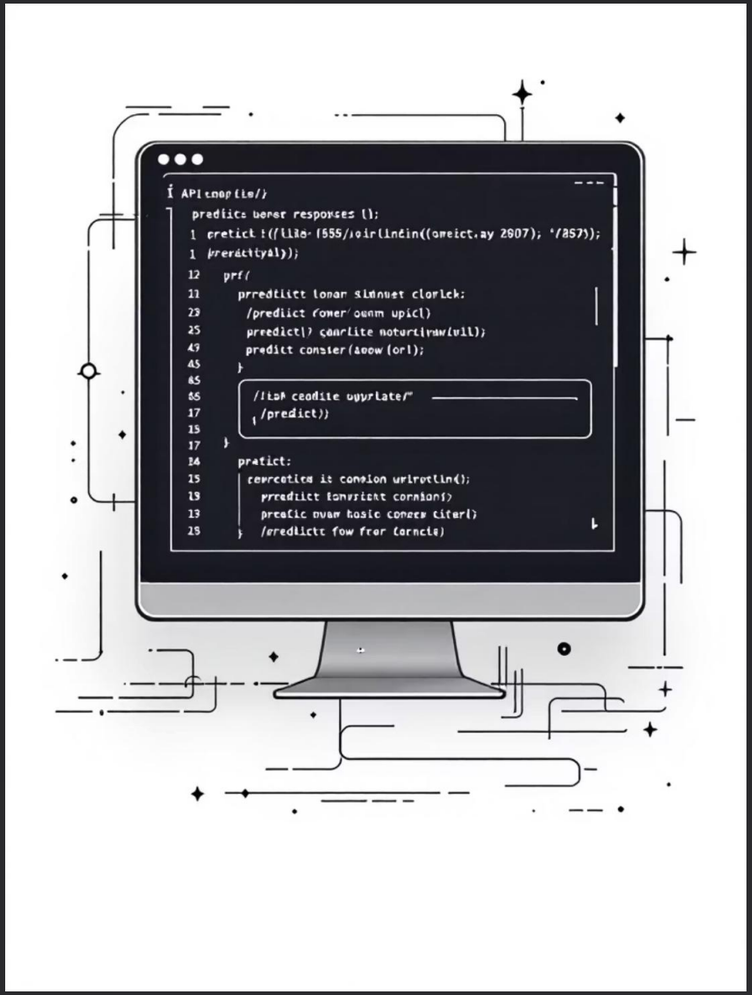
ML

Scikit-learn · TF-IDF · Regression model for continuous scoring

Machine Learning Approach

1. **Dataset:** Resumes labeled with ATS-like scores (synthetic & curated).
2. **Preprocessing:** PDF → parsed text, normalization, tokenization, stopword removal.
3. **Features:** TF-IDF vectors plus engineered features (section counts, keywords).
4. **Model:** Regression (scikit-learn) predicting continuous score 0–100 with cross-validation and error metrics.





Implementation — Backend & Frontend

- **FastAPI:** /predict endpoint serves model predictions and suggestions as JSON.
- **React UI:** Handles PDF upload, shows ATS Match %, rating breakdown, and inline suggestions.
- **Integration:** Asynchronous uploads, streaming parsing, and near real-time results for responsive user experience.

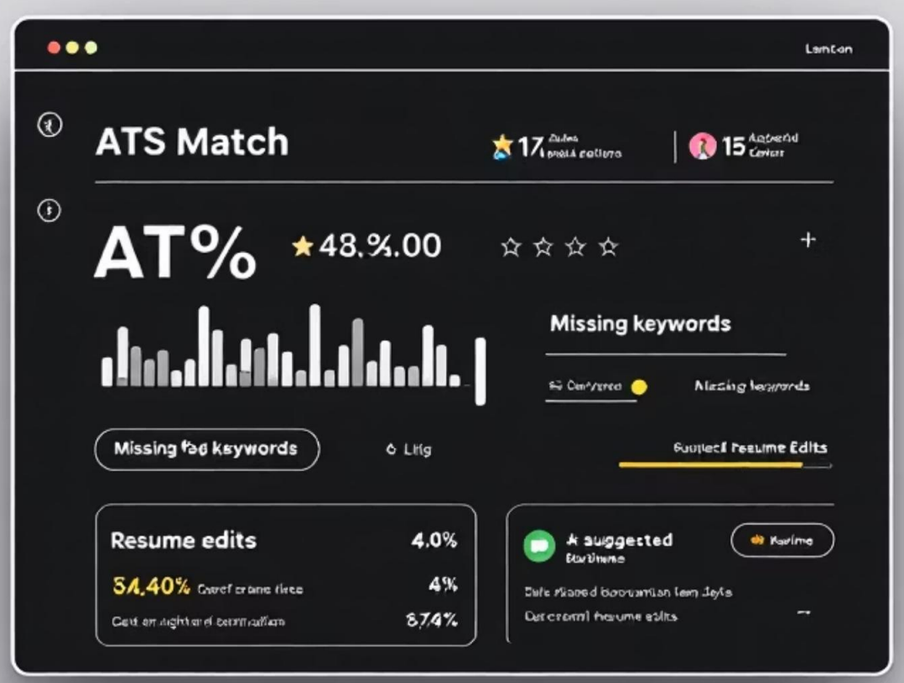
Results & Advantages

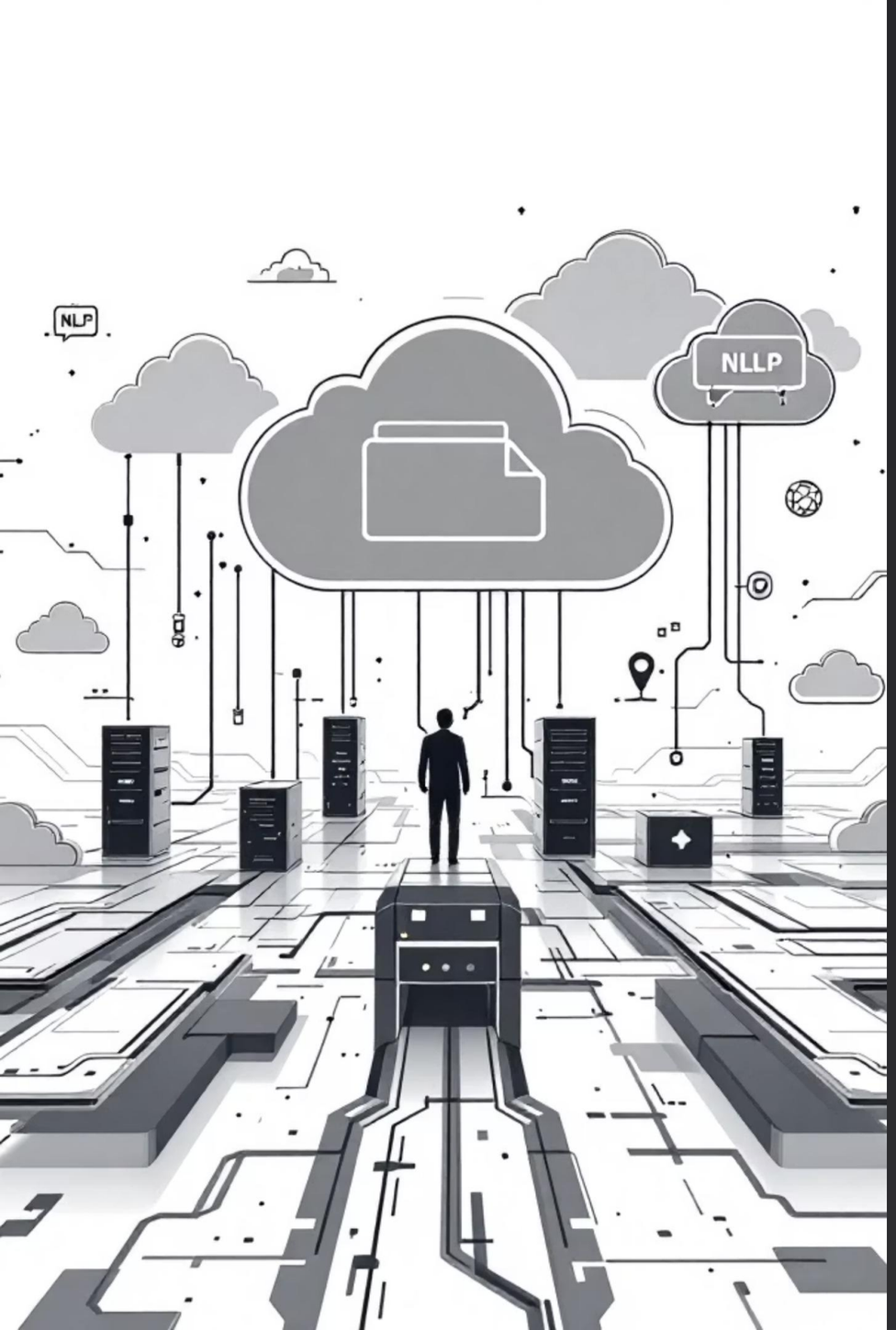
Example Outputs

ATS Match % · Resume Rating · General ATS Score (0–100) · Highlighted keywords

Advantages

Real-time feedback, ML-based continuous scoring, interpretable suggestions, improves candidate readiness for ATS & recruiters.





Conclusion & Future Scope

- **Impact:** Empowers candidates with actionable, data-driven resume feedback and streamlines recruiter screening.
- **Future Work:** Integrate advanced NLP (transformers), support more file formats, expand datasets, and cloud deployment with autoscaling.
- **Thank you.**